

ZUERICH-AFFOLTERN, Switzerland

WMO: 066640

Lat: 47.4278N

Lon: 8.5178E

Elev: 441

StdP: 96.14

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-6.3	-11.5	1.5	-7.1	-9.2	1.8	-5.0	11.0	8.3	9.4	8.3	1.6	50	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	31.1	20.0	29.0	19.3	27.1	18.8	21.0	28.4	20.1	26.9	19.4	25.5	2.2	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	14.2	23.2	17.9	13.6	22.4	17.2	13.0	21.7	62.5	28.5	59.6	27.0	57.0	25.8	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	6.4	5.2	-11.8	33.8	3.1	1.6	-14.0	35.0	-15.9	35.9	-17.6	36.8	-19.9	38.0		
			-12.2	22.8	3.4	0.9	-14.6	23.5	-16.6	24.0	-18.5	24.5	-20.9	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	1.2	1.7	5.8	10.0	14.0	18.3	19.7	19.1	15.0	10.7	5.3	1.8
	DBStd	7.64	4.12	4.24	3.96	3.60	3.58	3.58	3.14	3.15	3.24	3.72	3.55	3.77
	HDD10.0	1137	275	232	137	44	7	0	0	0	2	39	146	255
	HDD18.3	3126	533	465	389	251	142	45	22	28	109	238	391	513
	CDD10.0	1227	1	0	6	44	129	249	301	283	151	59	4	0
	CDD18.3	175	0	0	0	0	6	44	64	53	8	0	0	0
	CDH23.3	1644	0	0	0	8	87	432	597	459	60	1	0	0
	CDH26.7	497	0	0	0	1	13	131	197	151	4	0	0	0
Wind	WSAvg	1.7	2.0	2.0	2.1	1.8	1.7	1.5	1.4	1.3	1.2	1.2	1.6	2.0
Precipitation	PrecAvg	1207	69	72	75	88	131	136	140	132	97	88	90	89
	PrecMax	1541	170	162	227	175	243	222	257	236	208	159	195	185
	PrecMin	950	9	21	24	12	30	51	53	18	45	8	0	1
	PrecStd	154	42	38	44	43	58	44	54	52	40	40	49	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	14.6	19.7	24.4	28.5	32.7	33.6	33.4	27.1	22.0	16.8	12.0
		MCWB	9.3	8.6	11.1	13.7	18.0	20.8	20.3	20.0	18.5	15.7	11.9	8.9
	2%	DB	10.8	11.8	17.3	22.0	25.8	30.1	31.1	30.5	25.3	19.6	14.0	10.3
		MCWB	7.8	7.3	9.9	12.8	16.8	19.8	19.8	20.2	17.9	14.8	10.9	8.1
	5%	DB	8.9	9.9	15.0	19.9	23.5	28.0	28.8	28.1	22.9	17.9	12.0	8.8
		MCWB	6.5	6.5	9.0	11.9	15.4	19.1	19.2	19.4	16.8	14.1	9.9	6.9
	10%	DB	7.0	7.8	12.8	17.6	21.3	25.8	26.8	25.7	20.9	16.1	10.5	7.3
		MCWB	5.2	5.1	8.2	11.0	14.5	18.2	18.6	18.8	16.1	13.2	8.5	5.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	9.2	12.0	14.7	18.9	22.2	22.0	21.8	19.7	16.6	12.5	9.7
		MCDB	12.1	12.8	18.1	22.5	26.4	30.0	30.3	28.8	25.3	20.2	15.6	11.2
	2%	WB	8.2	7.7	10.6	13.4	17.6	20.7	20.8	20.8	18.6	15.6	11.2	8.3
		MCDB	10.3	11.2	15.3	20.2	24.0	27.7	28.4	27.9	23.6	18.6	13.5	10.0
	5%	WB	6.6	6.5	9.6	12.5	16.3	19.7	19.9	20.0	17.5	14.6	10.0	7.1
		MCDB	8.6	9.6	14.1	18.6	21.6	26.2	26.7	26.5	21.8	17.2	12.0	8.6
	10%	WB	5.2	5.3	8.5	11.5	15.1	18.8	19.1	19.1	16.5	13.6	8.7	5.8
		MCDB	6.9	7.6	12.5	16.9	19.9	24.6	25.1	24.8	20.0	15.7	10.2	7.1
Mean Daily Temperature Range	5% DB	MDBR	5.2	7.0	9.4	10.6	10.2	10.6	10.8	10.3	9.4	7.8	5.8	4.9
		MCDBR	7.8	11.4	14.4	15.7	14.8	14.6	15.0	14.6	12.7	10.7	8.7	6.7
		MCWBR	5.4	7.1	7.9	7.6	6.7	5.7	5.5	5.6	6.1	6.2	5.8	4.8
	5% WB	MCDBR	7.1	10.4	12.6	14.0	12.7	13.3	13.5	13.2	11.4	9.6	8.1	6.1
		MCWBR	5.1	6.8	7.1	6.9	5.9	5.4	5.2	5.4	5.6	5.9	5.5	4.8
Clear-Sky Solar Irradiance	taub	0.295	0.320	0.356	0.390	0.404	0.408	0.410	0.402	0.377	0.351	0.319	0.291	
	taud	2.444	2.373	2.330	2.269	2.275	2.303	2.292	2.329	2.382	2.437	2.477	2.453	
	Ebn at Noon	795	842	860	862	861	857	849	840	829	795	757	753	
	Edh at Noon	66	87	107	125	129	126	126	116	101	82	64	59	
All-Sky Solar Radiation	RadAvg	1.17	2.01	3.22	4.33	4.98	5.72	5.54	4.83	3.62	2.28	1.27	0.96	
	RadStd	0.13	0.27	0.43	0.62	0.50	0.53	0.51	0.50	0.37	0.25	0.15	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.44	N/A	N/A	N/A	N/A	N/A
(47)	Regional (117 neighbors)	N/A	N/A	N/A	+0.70	+0.42	+0.34	N/A	N/A	+56	+15

Nomenclature: See separate page